CHE210: ***DISCUSSION SESSION 12*** **NAME:**___________________________________

NOVEMBER 19, 2019

Ethylene oxide is produced by the catalytic oxidation of ethylene:

$$C_2H_{4\,(g)} + \frac{1}{2}O_{2\,(g)} \rightarrow C_2H_4O_{(g)}$$

An undesired competing reaction is the combustion of ethylene to CO_2.

The feed to a reactor contains 2 mol C_2H_4/mol O_2. The conversion and yield in the reactor are respectively 25% and 0.70 mol C_2H_4O produced/mol C_2H_4 consumed. A multiple-unit process separates the reactor outlet stream components: C_2H_4 and O_2 are recycled to the reactor, C_2H_4O is sold, and CO_2 and H_2O are discarded. The reactor inlet and outlet streams are each at 450°C, and the fresh feed and all the species leaving the separation process are at 25°C. The combined fresh feed-recycle stream is preheated to 450°C.

a) Taking the basis of 2 mol of ethylene entering the reactor, draw and label a flowchart of the complete process (show the separation process as a single unit) and calculate the molar amounts of all the process streams.
b) Calculate the heat requirement (kJ) for the entire process and that for the reactor alone. Use the tables below, filling in any missing entries.
c) Calculate the flow rate (kg/day) of the fresh feed and the overall and reactor heat requirements (kW) for a production rate of 1500 kg C_2H_4O/day.

Useful Data:

$$\Delta H^\circ_{f,C_2H_4O} = -51.00\,\frac{kJ}{mol}$$

$$\Delta H^\circ_{f,C_2H_4} = 52.28\,\frac{kJ}{mol}$$

$$C_{p,C_2H_4O}[J/(mol\cdot K)] = -4.69 + 0.2061\,T - 9.995\times10^{-5}\,T^2$$

$$C_{p,C_2H_4}[kJ/(mol\cdot ℃)] = 40.75\times10^{-3} + 11.47\times10^{-5}\,T - 6.897\times10^{-8}\,T^2 + 17.66\times10^{-12}\,T^3$$

OVERALL PROCESS:

Substance	n_{in} (mol)	$\hat{H}_{in}$ (kJ/mol)	n_{out} (mol)	$\hat{H}_{out}$ (kJ/mol)
C_2H_4			-	-
O_2		0	-	-
C_2H_4O	-	-		
CO_2	-	-		-393.5
$H_2O_{(l)}$	-	-		-285.84

ACROSS REACTOR:

Substance	n_{in} (mol)	$\hat{H}_{in}$ (kJ/mol)	n_{out} (mol)	$\hat{H}_{out}$ (kJ/mol)
C_2H_4				
O_2		13.37		13.37
C_2H_4O	-	-		
CO_2	-	-		-374.66
$H_2O_{(g)}$	-	-		-226.72